Corporate Finance - Problem Set 1

Exercise 1 Answer the following questions.

(a) Why do we need to discount free cash flows (FCFs) instead of a ready-to-use accounting figure, such as net income? State and explain the differences between net income and free cash flows.

(b) Taxes that accrue in one period are not necessarily paid in the same period. As a result, income taxes — as recorded in the income statement — may differ from the amount actually paid in taxes in the current period. This difference is recorded as *deferred tax* liability in the balance sheet. So income taxes recorded on the income statement = taxes paid (cash) + changes in deferred taxes. In real world applications, should we adjust the textbook FCF formula, $FCF_t = EBIT_t(1-\tau_c) + Depreciation - CAPEX_t - \Delta NWC_t$, to include changes in deferred taxes?

Exercise 2 You are considering an investment of 25 million in a new equipment. You pay a consultancy company 1 million to help you with the analysis. You open the consultants' report and find the following estimates (in thousands of dollars) for the project:

	1	2	3	4	5	6	7	8	9	10
Sales revenue	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000
Costs of gods sold	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000
Gross profit	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000
Overhead	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000
Depreciation	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500
Net operating income	7,500	7,500	7,500	7,500	7,500	7,500	7,500	7,500	7,500	7,500
Income tax	2625	2625	2625	2625	2625	2625	2625	2625	2625	2625
Net income	4,875	4,875	4,875	4,875	4,875	4,875	4,875	4,875	4,875	4,875

All of the estimates in the report seem correct. You note that the consultants used straight line depreciation for the new equipment that will be purchased today (year 0), which is what the accounting department recommended. The project will require 10 million in working capital upfront (year 0), which will be fully recovered in year 10. Next, you see they have attributed 2 million of overhead to the project but you know that 1 million of this amount will be incurred even if the project is not accepted.

(a) Given the available information, what are the free cash flows in years 0 through 10 that should be used to evaluate the proposed project?

(b) If the cost of capital for this project is 14%, what is your estimate of the value of the new project?

Exercise 3 One year ago, your company purchased a machine in manufacturing for 110,000. You have learned that a new machine is available that offers many advantages; you can purchase it for 150,000 today. It will be depreciated on a straight-line basis over 10 years, after which it has no salvage value. You expect that the new machine will produce additional EBITDA (Earnings before interest, taxes, depreciation and amortization) of $40,000 per year for the next 10 years. The current machine is expected to produce

additional EBITDA of 20,000 per year. The current machine is being depreciated on a straight-line basis over a useful life of 11 years, after which it will have no salvage value, so depreciation expenses for the current machine is 10,000 per year. All other expenses of the two machines are identical. The market value today of the current machine is 50,000. Your company's tax rate is 45%, and the opportunity cost of capital for this type of equipment is 10%. Is it profitable to replace the year-old machine?